EXPERIMENT 3

AIM-To perform various GIT operations on local and Remote repositories.

THEORY:

Git is a free and open source distributed version control system designed to handle everything from small to very large projects with speed and efficiency. Git is easy to learn and has a tiny footprint with lightning fast performance. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like cheap local branching, convenient staging areas, and multiple workflows.

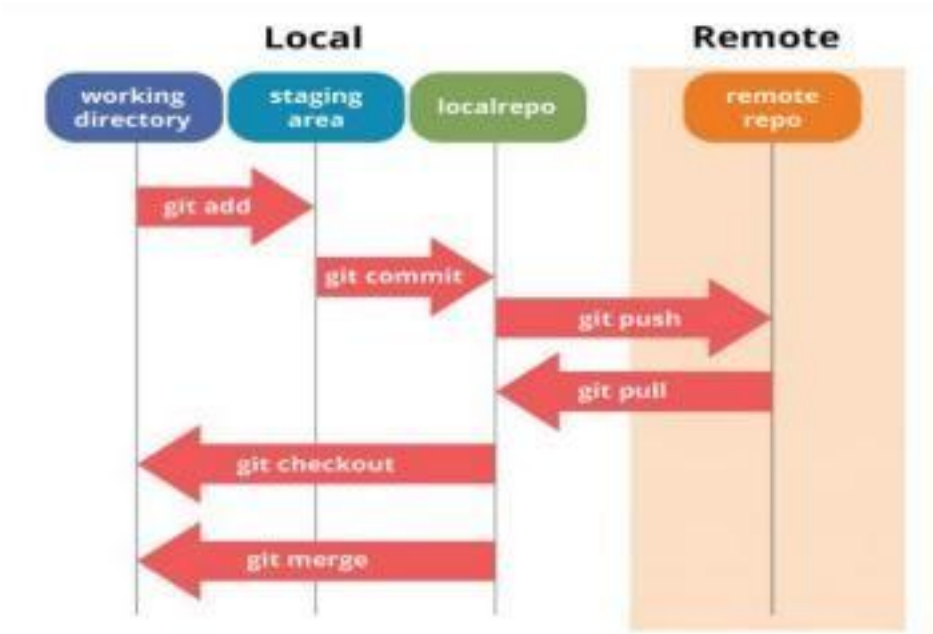
Some of the basic operations in Git are:

1. Initialize
2. Add
3. Commit
4. Pull
5. Push

Some advanced Git operations are:

1. Branching
2. Merging
3. Rebasing

The following diagram depict the all supported operations in GIT



Step 1: Initialize Git Repository (Only First Time)

Step 2: To add all files

Step 3: Commit the File

```
C:\Users\503\Desktop\T11_18>git init
Initialized empty Git repository in C:/Users/503/Desktop/T11_18/.git/

C:\Users\503\Desktop\T11_18>git add .

C:\Users\503\Desktop\T11_18>git status -v
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   index.py

diff --git a/index.py b/index.py
new file mode 100644
index 0000000..917eff0
--- /dev/null
+++ b/index.py
@@ -0,0 +1,3 @@
+# This is a simple Python program
+name = input("Enter your name: ")
+print("Hello,", name, "! Welcome to Python.")
```

Step 4: Connect to Remote Repository (Only First Time)

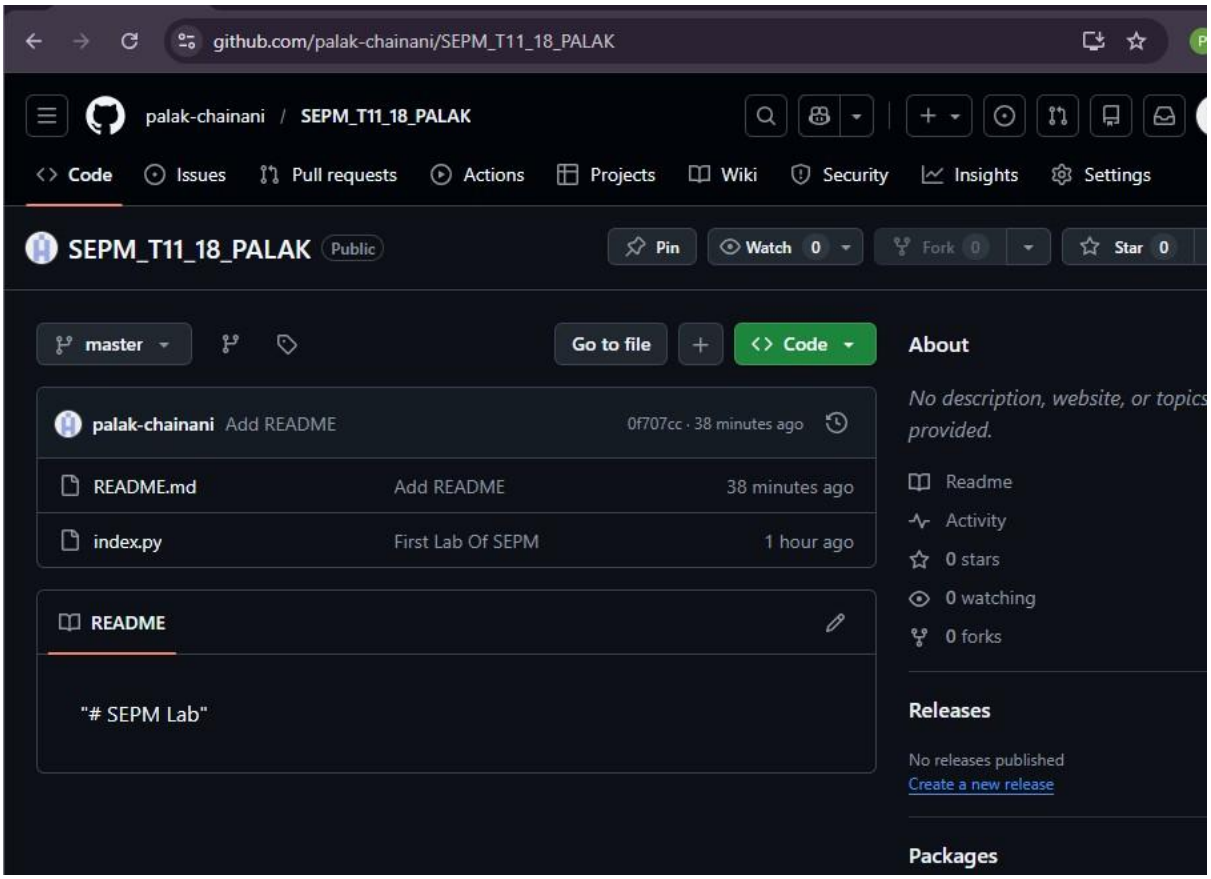
```
C:\Users\503\Desktop\T11_18>git remote add origin https://github.com/palak-chainani/SEPM_T11_18_PALAK.
git
```

Step 5: Push the Commit to GitHub

Upload the committed changes to GitHub.

```
C:\Users\503\Desktop\T11_18>git push -u origin master
fatal: User canceled device code authentication
Username for 'https://github.com': palak-chainani
Password for 'https://palak-chainani@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/palak-chainani/SEPM_T11_18_PALAK.git/'
```

ALL FILES ARE UPLOADED



CONCLUSION: Performed various git operations.